

4. Design of Peak detection on a given ECG signal using Xilinx Vivado

Aim

To design and simulate a Verilog-based ECG peak detection system that reads fixed-point ECG samples from a text file and identifies peak points above a predefined threshold using a digital comparator and local-maximum logic.

Software Required

- Xilinx Vivado Design Suite
- Vivado Simulator (XSIM)
- ECG file

Procedure

1. Prepare ECG Input File
 - Create a text file named ECG.txt.
 - Store fixed-point ECG samples in it, one value per line.
2. Create a New Project
 - Open Vivado and create a new RTL project.
 - Select the target FPGA device/board.
 - Add a new Verilog source file and write the `ecg_peak_detector` module.
 - Add another Verilog file for the testbench.
3. Design the Peak Detector
 - Implement a 3-sample shift register to store previous, current, and next ECG samples.
 - Compare the current sample with the previous and next samples.
 - Check whether the current sample is greater than a predefined threshold.
 - Generate a peak output when all conditions are satisfied.
4. Testbench Implementation
 - Use `$fopen` and `$fscanf` to read ECG samples from ECG.txt.
 - Apply one ECG sample to the detector at every clock cycle.
 - Monitor the peak signal in the simulation waveform and console.
5. Simulation
 - Set the testbench as the top module.
 - Run Behavioral Simulation.
 - Observe ECG waveform and peak pulses in the Vivado waveform window.

Peak detection program

```
module ecg_peak_detector #(
    parameter WIDTH = 16,
    parameter THRESHOLD = 16'sd1000
)(
    input wire          clk,
    input wire signed [WIDTH-1:0] ecg_in,
    output reg          peak
);
    reg signed [WIDTH-1:0] prev, curr, next;
    always @(posedge clk) begin
        // Shift register for 3 samples
        prev <= curr;
        curr <= next;
        next <= ecg_in;
        // Peak detection condition
        if ((curr > THRESHOLD) && (curr > prev) && (curr > next))
            peak <= 1'b1;
        else
            peak <= 1'b0;
        end
    endmodule
```

Testbench program

```
`timescale 1ns/1ps

module tb_ecg_peak;

    reg clk;

    reg signed [15:0] ecg_in;

    wire peak;
```

```

integer fd_ecg;

integer r;

// Clock generation
always #5 clk = ~clk;

// DUT
ecg_peak_detector #(
    .WIDTH(16),
    .THRESHOLD(16'd45000)
) dut (
    .clk(clk),
    .ecg_in(ecg_in),
    .peak(peak)
);

initial begin
    clk = 0;
    ecg_in = 0;

    // Open ECG file
    fd_ecg = $fopen("I:/Saveetha/SOC/Lab experiments/ECG.txt", "r");

    // Read ECG samples
    while (!$feof(fd_ecg)) begin
        r = $fscanf(fd_ecg, "%d\n", ecg_in);

        @(posedge clk);

        if (peak)
            $display("Peak detected at time %t, ECG = %d", $time, ecg_in);
    end

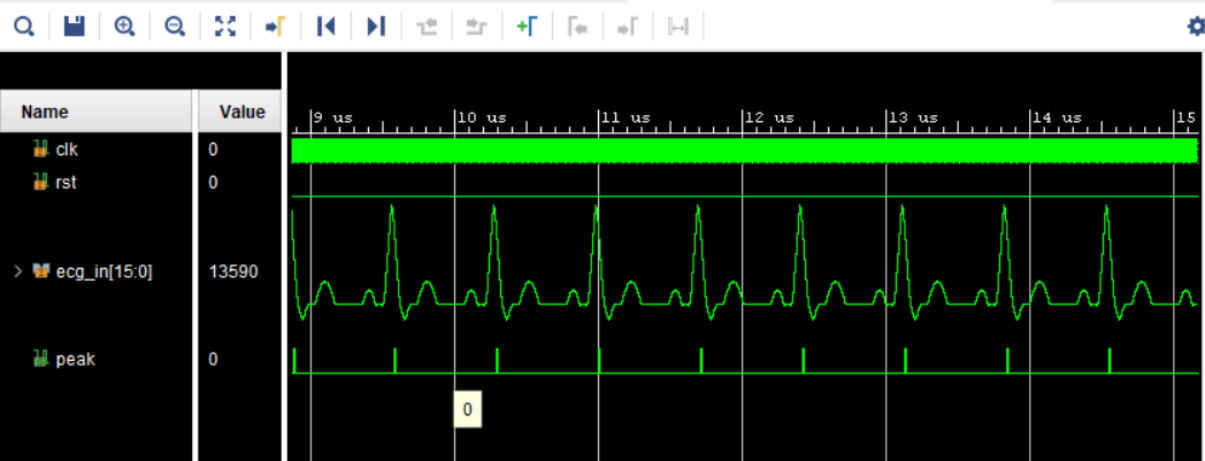
    $fclose(fd_ecg);

    #20 $finish;
end

```

endmodule

Output waveform



Result

The ECG peak detection system was successfully designed and simulated in Vivado. Fixed-point ECG samples were read from the text file and processed sequentially. Whenever the ECG signal exceeded the preset threshold and formed a local maximum, the peak signal was asserted. The waveform results confirmed correct identification of ECG peak points, verifying the proper operation of the Verilog-based ECG peak detector.